Amplitude Modulation(AM)

Amplitude modulation :

- **Amplitude of the sinusoidal carrier signal is varied in accordance with the amplitude of the baseband signal**
  - Frequency and phase not altered

Types of Amplitude Modulation (AM)

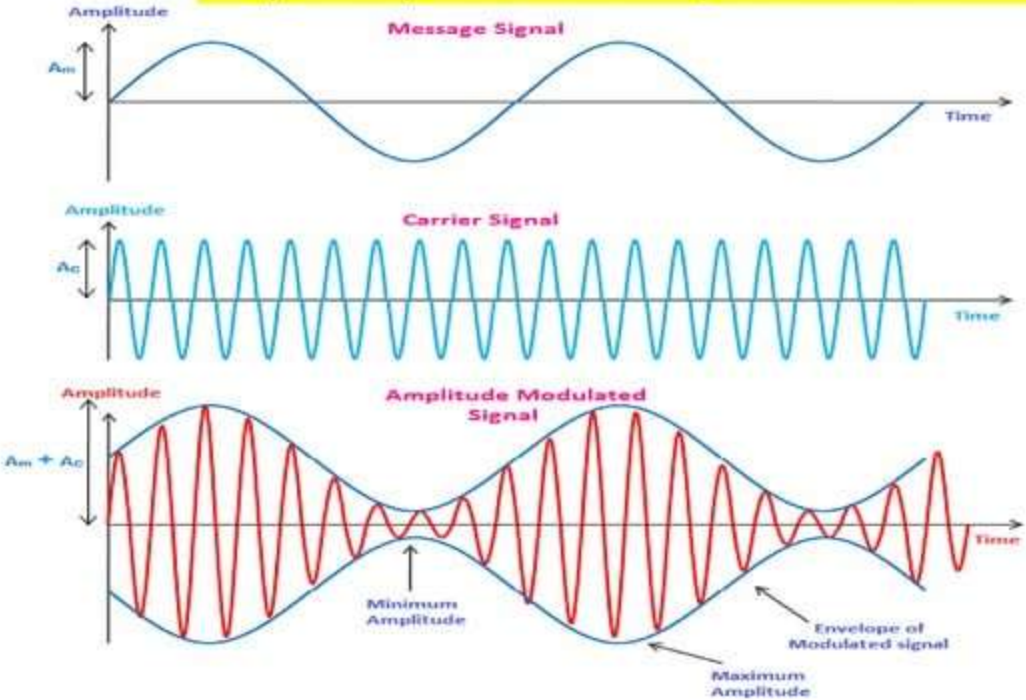
- **Double Sideband with carrier (we will call it AM)**: This is the most widely used type of AM modulation. In fact, all radio channels in the AM band use this type of modulation.
- **Double Sideband Suppressed Carrier (DSBSC):** This is the same as the AM modulation above but without the carrier.
- **Single Sideband (SSB):** In this modulation, only half of the signal of the DSBSC is used.
- **Vestigial Sideband (VSB):** This is a modification of the SSB to ease the generation and reception of the signal.

Amplitude modulation

Double sideband with full carrier(DSBFC)
(conventional AM, standard AM)

- Amplitude of the sinusoidal carrier signal is varied in accordance with the amplitude of the baseband signal.
- AM wave consists of carrier

Graphical representation of Amplitude modulation



"standard" or double sideband, full carrier(DSBFC) **AM** (officially known as A3E modulation)

Mathematical model of AM

Amplitude of carrier

Let the **carrier wav** $v_c = V_c \sin \omega_c t$ $\omega_c = 2\pi f_c$-carrier frequency

modulating signal $v_m = V_m \sin \omega_m t$

Amplitude of modulating signal

- amplitude V_c of the unmodulated carrier will have to be made proportional to the *instantaneous* modulating voltage $V_m \sin \omega_m t$
  - **amplitude of the amplitude-modulated(AM) wave**

$$A = V_c + v_m = V_c + V_m \sin \omega_m t$$
$$= V_c (1 + m \sin \omega_m t) \quad \text{---(1)} \qquad \text{where } m = \frac{V_m}{V_c}$$

m -modulation index(depth of modulation) (0 to 1)

modulation index : $m = \frac{V_m}{V_c}$ $= K_a V_m$ $V_m < V_c$ → To avoid distortion

Ratio of maximum amplitude of the modulating signal to the maximum amplitude of the carrier wave

Amplitude of modulating wave

Amplitude sensitivity($1/V_c$)

percentage modulation.

$$m \times 100 = \quad \times 100$$

m =0.5 means 50% modulation

Degree of modulation

If > 1 → **over modulation** of carrier occurs

resulting in carrier phase reversals

envelope distortion occurs

If = 1 → **critical modulation** of carrier occurs

If < 1 → **under modulation** of carrier occurs

Under modulated (<100%)

100% modulated

Over Modulated (>100%)

Envelope Detector
Can be used

Envelope Detector
Gives Distorted signal

Instantaneous voltage of the AM wave v is

Using (1)

$$v = A \sin \omega_c t = V_c (1 + m \sin \omega_m t) \sin \omega_c t \quad \text{-----} (2)$$

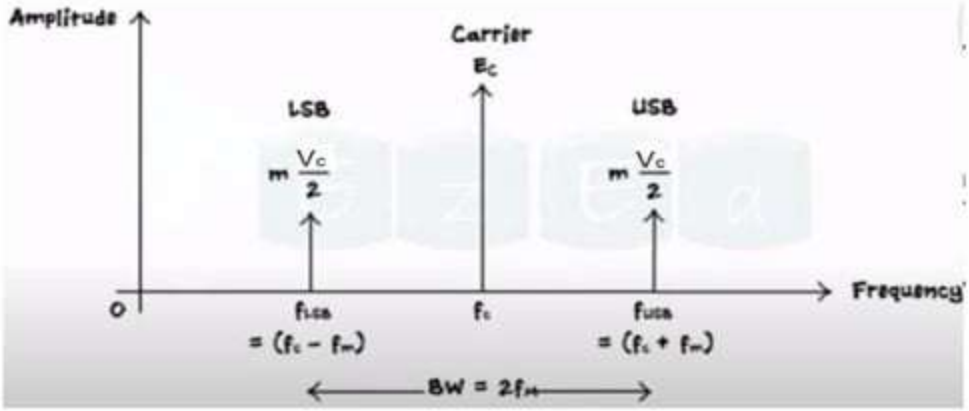
using $\sin x \sin y = ½ [\cos (x - y) - \cos (x + y)]$

$$v = V_c \sin \omega_c t + \frac{mV_c}{2} \cos (\omega_c - \omega_m)t - \frac{mV_c}{2} \cos (\omega_c + \omega_m)t \quad \text{---}(3)$$

unmodulated carrier

lower sideband (LSB) $f_c - f_m$

frequency of the upper sideband (USB) $f_c + f_m$



Spectrum of AM wave

DSBFC modulated wave is

$$x(t)\cos\omega_c t \leftrightarrow \frac{1}{2}\left[X(\omega-\omega_c)+X(\omega+\omega_c)\right]$$

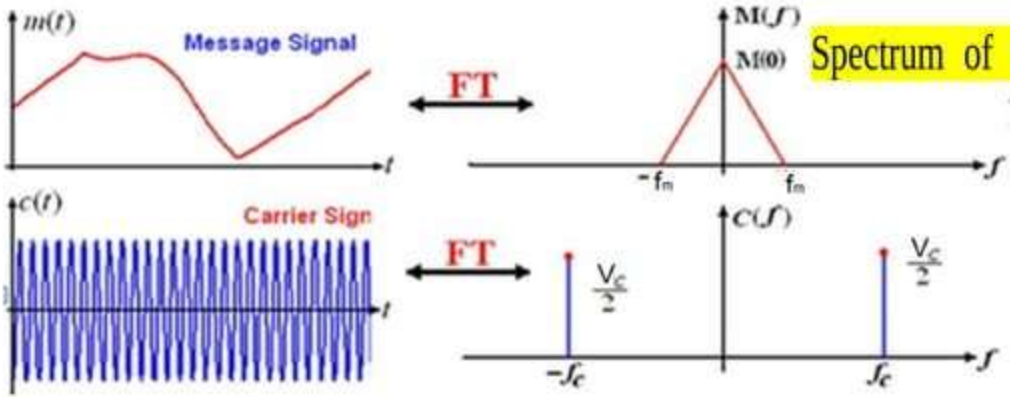
$$v = V_c\,(1 + m \sin \omega_m t)\ \sin \omega_c t$$

$$v = V_c \sin \omega_c t + K_a V_c\, m(t)\ \sin \omega_c t$$

is obtained by taking the

Fourier transform of the AM wave

$$V(f) = \frac{V_c}{2}[\delta(f - f_c) - \delta(f + f_c)] + \frac{K_a V_c}{2}[M(f - f_c) - M(f + f_c)]$$

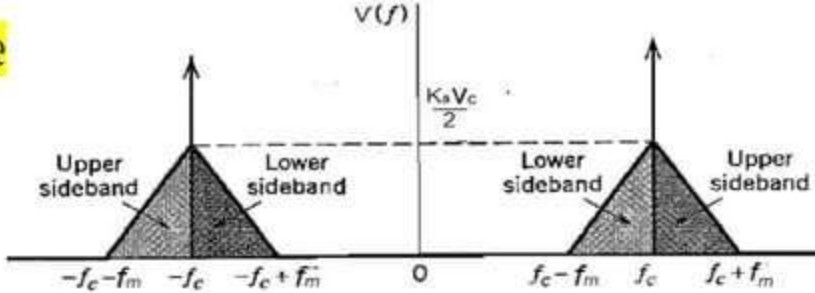


Spectrum of baseband signal

band-limited

($-f_m$ to f_m)

Spectrum of AM wave



Spectrum of AM wave consists of

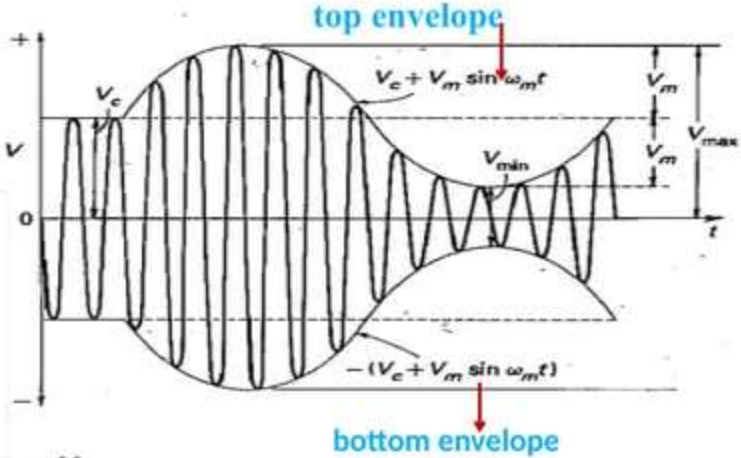
1. **two delta functions with weight $V_c/2$ occurring at $\pm f_c$**
2. **two versions of baseband frequency translated in frequency by $\pm f_c$ and scaled by $K_aV_c/2$**

- **For +ve freq. spectrum above f_c , upper side band. Below fc lower side band**
- **For - ve freq. spectrum below f_c , upper side band. above f_c lower side band**
- **central frequency(carrier) -highest amplitude, other two(USB,LSB) frequencies symmetrical about centre and equal amplitude, less than carrier amplitude**

Bandwidth of AM = $(f_c + f_m) - (f_c - f_m) = 2f_m$ (twice the highest modulating frequency)

Modulation index in terms of max, min voltage of AM wave

- Amplitude of the carrier is modulated by message
- Envelope of the carrier contains the information about message



$$V_m = \frac{V_{max} - V_{min}}{2}$$

$$V_c = V_{max} - V_m = V_{max} - \frac{V_{max} - V_{min}}{2}$$

$$= \frac{V_{max} + V_{min}}{2}$$

$$m = \frac{V_m}{V_c} = \frac{(V_{max} - V_{min})/2}{(V_{max} + V_{min})/2}$$

$$\boxed{m = \frac{(V_{max} - V_{min})}{(V_{max} + V_{min})}}$$

Power Relations in the AM Wave

total power in the modulated wave

$$P_t = \frac{V_{carr}^2}{R} + \frac{V_{LSB}^2}{R} + \frac{V_{USB}^2}{R} \text{ (rms)}$$

unmodulated -carrier power is

From(2)

$$P_c = \frac{V_{carr}^2}{R} = \frac{(V_c/\sqrt{2})^2}{R} = \frac{V_c^2}{2R}$$

Sideband power is

$$P_{LSB} = P_{USB} = \frac{V_{SB}^2}{R} = \left(\frac{mV_c/2}{\sqrt{2}}\right)^2 \div R = \frac{m^2 V_c^2}{8R}$$

Therefore, total power is

$$P_t = \frac{V_c^2}{2R} + \frac{m^2}{4}\frac{V_c^2}{2R} + \frac{m^2}{4}\frac{V_c^2}{2R} = P_c + \frac{m^2}{4}P_c + \frac{m^2}{4}P_c$$

$$\frac{P_t}{P_c} = 1 + \frac{m^2}{2}$$

maximum power in the AM wave is $P_t = 1.5P_c$ when m = I.

Transmission efficiency

Ratio of power in side band to total power

$\% \eta$ = (power in side band/ total power) x 100

$$\% \eta = P_{SB}/P_t = (P_c m^2/2) / (P_c + P_c m^2/2)$$

$$= (m^2/2) / (1+m^2/2)$$

If m=1, $\% \eta = 1/3 = 33.3\%$

$$\% \eta = 33.3\%$$

Only 33% of power is used for transmission
Remaining power is wasted in the carrier transmission along with sidebands

Modulation by several sine waves(modulation signal)

Let *V1*, *V2* , V3, etc., be the simultaneous modulation voltages

Total modulating voltage V_t

$$V_t = \sqrt{V_1^2 + V_2^2 + V_3^2 + \cdots}$$

$$\frac{V_t}{V_c} = \frac{\sqrt{V_1^2 + V_2^2 + V_3^2 + \cdots}}{V_c}$$

$$= \sqrt{\frac{V_1^2}{V_c^2} + \frac{V_2^2}{V_c^2} + \frac{V_3^2}{V_c^2} + \cdots}$$

$$m_t = \sqrt{m_1^2 + m_2^2 + m_3^2 + \cdots}$$

total power

$$P_t = P_c\left(1 + \frac{m^2}{2}\right) = P_c + \frac{P_c m^2}{2} = P_c + P_{SB}$$

Carrier power will be unaffected, but the total sideband power

$$P_{SB_T} = P_{SB_1} + P_{SB_2} + P_{SB_3} + \cdots$$

$$\frac{P_c m_t^2}{2} = \frac{P_c m_1^2}{2} + \frac{P_c m_2^2}{2} + \frac{P_c m_3^2}{2} + \cdots$$

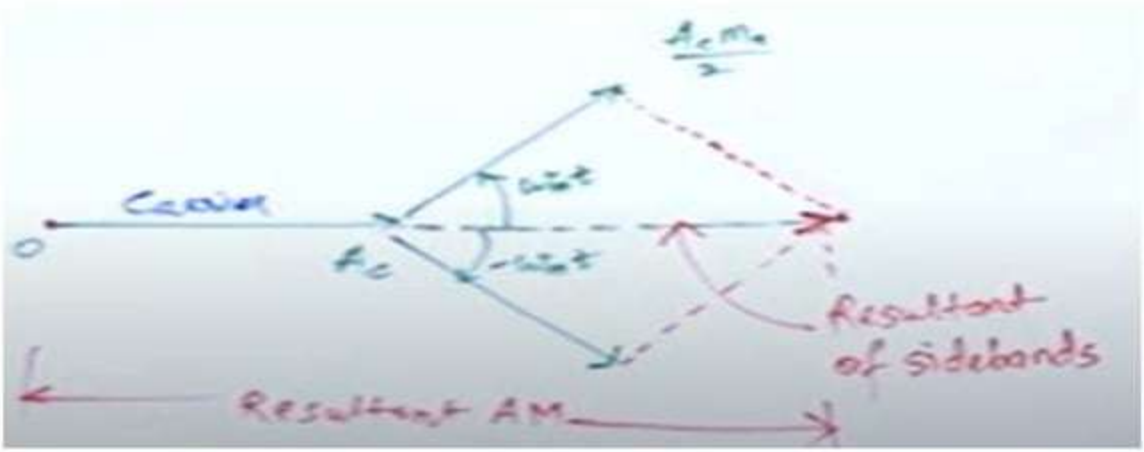
$$m_t^2 = m_1^2 + m_2^2 + m_3^2 + \cdots$$

Phase spectrum

Phasor representation of AM

The carrier is taken as the reference phasor and the two sideband phasors are rotating in the opposite direction.

The resultant phasor is the sum of carrier phasor and two sideband phasors rotating in the opposite direction.



Disadvantages

- Carrier conveys no information
- two sidebands are images of each other, All the information can be conveyed by the use of one sideband
- Wasteful of power (carrier wave is not necessary)
- Wasteful of bandwidth (two side bands)

Advantages

- used for broadcasting
- simplicity of the modulating and demodulating equipment

AMPLITUDE MODULATORS(AM Generation)

Nonlinear modulators

- It makes use of the non linear characteristics of nonlinear device. (diode, transistor, FET)
- To make the device to operate in the nonlinear VI characteristics, the input is kept low.

Linear modulator:

- It makes use of the linear region of the VI characteristics of diode
- The input is kept high

Methods

1. square-Law Modulation
2. Switching Modulator

1. Square-Law Modulation-nonlinear

- Square law diode modulation makes use of non-linear current-voltage characteristics of diode.
- This method is suited for low voltage levels

It consists of

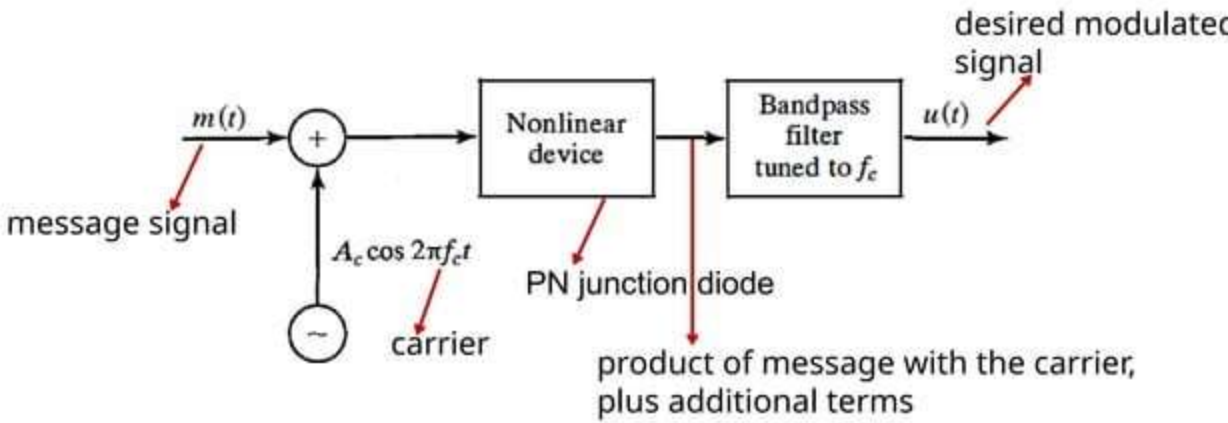
- A carrier source and modulating signal, A non-linear device and bandpass filter

The modulating signal and carrier are connected in series with each other and their sum $v_i(t)$ is applied at the input of the non-linear device, such as diode, transistor etc.

Principle of square law diode modulator

- when two different frequencies are simultaneously passed through a nonlinear device, such as diode, the process of amplitude modulation occurs.

- when carrier and modulating frequencies are simultaneously applied at the input of diode, different frequency terms appear at the output of diode.
- These different frequency terms are applied to the bandpass filter to extract the carrier frequency and two sidebands and reject other frequencies.

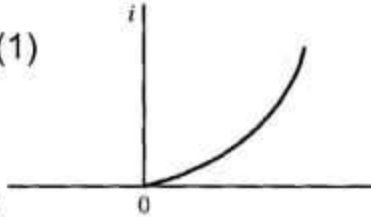


- AM modulator- consists of nonlinear device(switching modulator).
- message+ carrier(high amplitude) is applied to diode.
- AM component is extracted by means of bandpass filter from the output voltage of the nonlinear device.

Output of nonlinear device $v_o(t)$

input to the nonlinear device

$$v_0(t) = a_1 v_i(t) + a_2 v_i^2(t) \quad \text{-----(1)}$$



(a1, a2) are constants

input to the nonlinear device

$$v_i(t) = m(t) + A_c \cos 2\pi f_c t \quad \text{-----(2)}$$

output of the nonlinear device Substituting (2) in (1)

$$v_o(t) = a_1[m(t) + A_c \cos 2\pi f_c t] + a_2[m(t) + A_c \cos 2\pi f_c t]^2$$

$$= a_1 m(t) + a_1 A_c \cos 2\pi f_c t] + a_2 m^2(t) + a_2 A_c^2 \cos^2 2\pi f_c t + 2a_2 A_c m(t) \cos 2\pi f_c t]$$

$$= a_1 m(t) + a_2 m^2(t) + a_2 A_c^2 \cos^2 2\pi f_c t + A_c a_1 \left[1 + \frac{2a_2}{a_1} m(t)\right] \cos 2\pi f_c t.$$

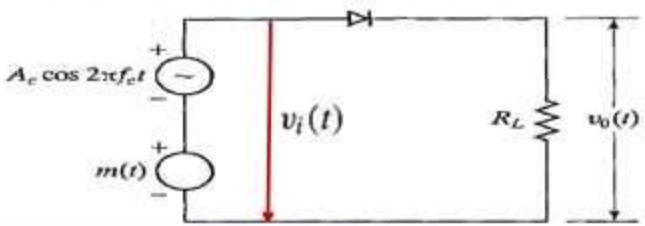
output of the bandpass filter with a bandwidth 2 W centered at f = fc yields

$$\boxed{u(t) = A_c a_1 \left[1 + \frac{2a_2}{a_1} m(t)\right] \cos 2\pi f_c t}$$

where $2a_2|m(t)|/a_1 < 1$

2. Switching Modulator- linear

Large signal



- Switching modulator is similar to the square law modulator.
- only difference is that in the square law modulator, the diode is operated in a non-linear mode, whereas, in the switching modulator, the diode has to operate as an ideal switch.

Let the modulating signal is $m(t)$

carrier signals is $c(t) = A_c \cos(2\pi f_c t)$

These two signals are applied as inputs of the diode.

Input to a diode is sum of message signal and the carrier signal

$$v_i(t) = m(t) + A_c \cos 2\pi f_c t$$

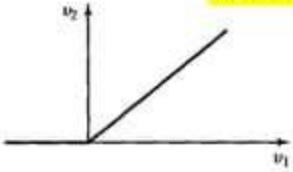
Assume, the magnitude of the modulating signal is very small when compared to the amplitude of carrier signal Ac
.So, the diode's ON and OFF action is controlled by carrier signal c(t).

diode will be forward biased when c(t)>0 and it will be reverse biased when c(t)<0

.

Diode characteristic

diode has linear input-output voltage characteristic where $A_c \gg m(t)$.



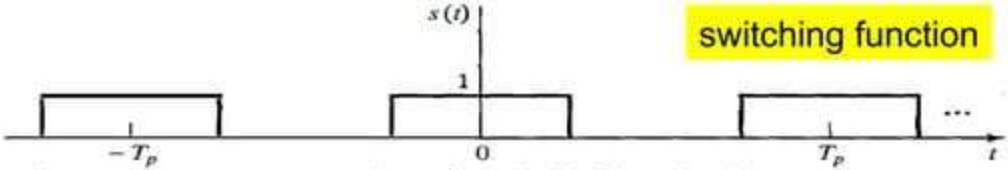
output of the diode is $v_0(t) = \begin{cases} v_i(t), & c(t) > 0 \\ 0, & c(t) < 0 \end{cases}$

We can approximate this as $v_0(t)=v_i(t)s(t)$

mathematically $v_0(t) = [m(t) + A_c \cos 2\pi f_c t]s(t)$

switching function

where, s(t) is a periodic pulse train with time period $T_p=1/f_c$



switching function

Fourier series representation of periodic function is

$$s(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}\cos[2\pi f_c t(2n-1)]$$

$$v_0(t) = [m(t) + A_c\cos 2\pi f_c t]s(t)$$

$$= \frac{A_c}{2}\left[1 + \frac{4}{\pi A_c}m(t)\right]\cos 2\pi f_c t + \text{other terms.}$$

$v_o(t)$ is passes through a bandpass filter with the center frequency $f = f_c$ and the bandwidth 2W to get the desired AM-modulated.

Desired conventional AM signal is

$$\boxed{u(t) = \frac{A_c}{2}\left[1 + \frac{4}{\pi A_c}m(t)\right]\cos 2\pi f_c t}$$

standard equation of AM wave is $s(t)=Ac[1+k_a m(t)]\cos(2\pi f_c t)$

comparing with the standard equation of AM wave, we will get the scaling factor as 0.5 and amplitude sensitivity $ka = 4/\pi A_c$.

AM Demodulation

-Envelope Detector

- process of extracting an original message signal from the modulated wave is known as **detection** or **demodulation**.
- The circuit, which demodulates the modulated wave is known as the **demodulator**.

- At the Receiver, AM demodulation is done by envelope detector
- Detector Consists of a **diode and low pass filter**.
- Diode is the main detecting element. Hence, the envelope detector is also called as the **diode detector**.
- The **low pass filter contains a parallel combination of the resistor and the capacitor**.

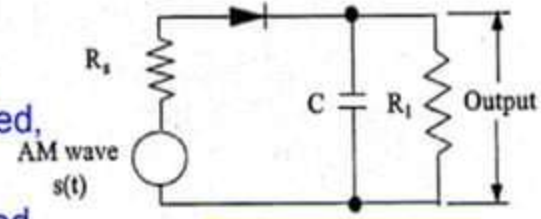
- When carrier frequency is high enough, and % of modulation is less than 100%, the demodulated output is same as envelope of the message signal

Demodulation of AM signals

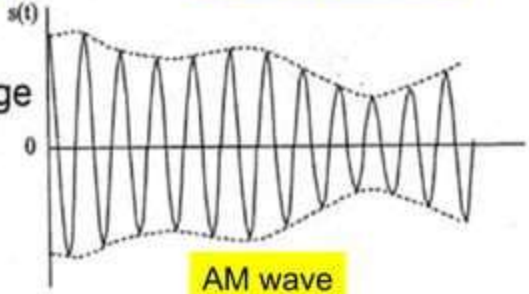
Envelop Detection

- On +ve half cycle, diode is forward-biased, capacitor is charged to peak value
- On –ve half cycle, diode is reverse-biased and capacitor discharges slowly through load resistor R_l.
- Assume AM wave was supplied by voltage source with internal impedance R_s.

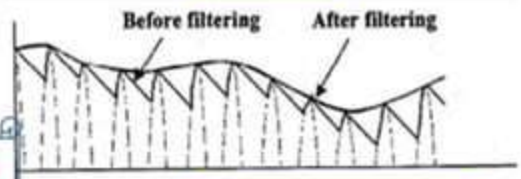
- When the value of AM wave is greater than the capacitor voltage, the diode conducts and the process will be repeated.



Envelop Detector



AM wave



Detected envelope

Envelop Detector is Simple and efficient when percentage modulation < 100%

capacitor voltage waveform is same as that of the envelope of AM wave

- RC is too small, output of the filter falls very rapidly after each peak and will not follow the envelope of the modulated signal
- **BW of LPF is too large**

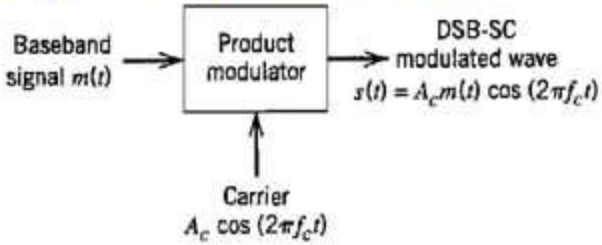
- If RC is too large, discharge of the capacitor is too slow and output will not follow the envelope of the modulated signal.
- **BW of LPF is too small**

For good performance

$$\frac{1}{f_c} \ll RC \ll \frac{1}{f_m}$$

Double-Sideband Suppressed-Carrier AM(DSB-SC)

Double-sideband suppressed-carrier (**DSB-SC**) modulation is an **amplitude modulation** that consists only of the two symmetrical **sidebands** and no **carrier** band.



Generated by product modulator

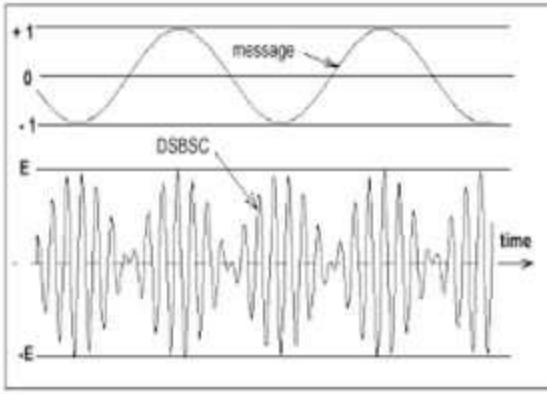
message signal = m(t)

Carrier signal $c(t) = A_c \cos(2nf_c t)$

DSB-SC AM signal

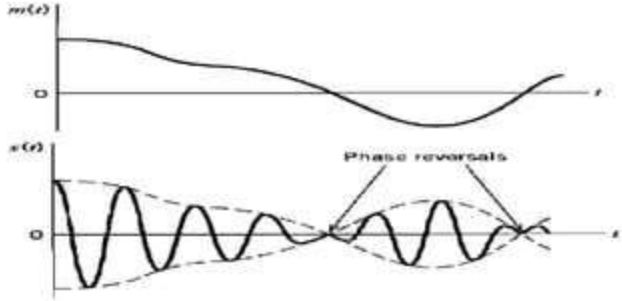
$$u(t) = m(t)c(t)$$

$$= A_c m(t) \cos(2\pi f_c t).$$



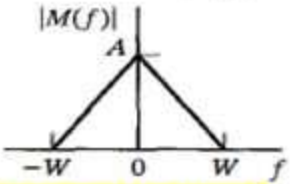
- slowly varying m(t) is changed into a rapidly varying u(t)
- u(t) contains higher-frequency components due to its rapid changes with time
- At the same time, u(t) retains the main characteristics of the message signal; therefore, it can be used to retrieve the message signal at the

- modulated signal undergo phase reversal whenever m(t) crosses zero

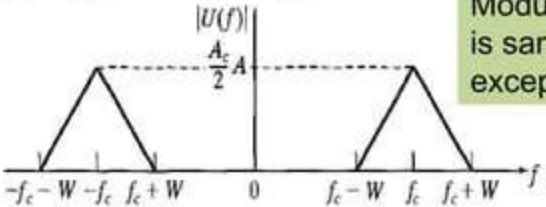


Spectrum of the DSB-SC

$$U(f) = \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]$$



spectra of m(t)



spectra of DSB-SC AM signal

Modulated signal spectrum is same as message except translation by ±fc

- No impulse in U(f) at f = f_c, since the carrier is absent. Therefore, u(t) is called a **suppressed-carrier signal**

upper sideband of U(f) contains all the frequency content of M(f) for f > 0
Lower sideband of U(f) contains all the frequency content of M(f) for f < 0

upper and lower sideband of U(f) contains all the frequency content of M(f)

Power Content of DSB-SC Signals

$$P_t = \frac{V_c^2}{2R} + \frac{m^2}{4}\frac{V_c^2}{2R} + \frac{m^2}{4}\frac{V_c^2}{2R}$$

power content of the modulated signal P_u

P_u = Upper sideband power + Lower sideband power

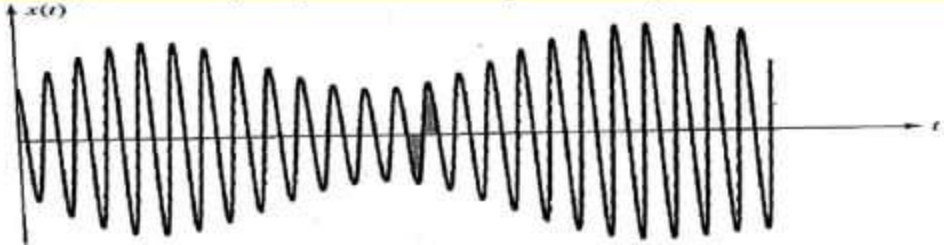
unmodulated -carrier power is $Pc=Ac^2/2R$

$$P_u = P_c m^2/4 + P_c m^2/2)$$
$$= P_c m^2/2$$

Sideband power is

$$P_{LSB} = P_{USB} = \frac{V_{SB}^2}{R} = \left(\frac{mV_c/2}{\sqrt{2}}\right)^2 \div R = \frac{m}{}$$

$m^2(t)$ is a slowly varying signal. When multiplied by high-frequency sinusoid $\cos(4nf_c t)$, results is a high-frequency sinusoid with a slowly varying envelope with equal positive & negative amplitude, cancel each other



Power Content of DSB-SC Signals

power content of the modulated signal P_u

P_u = Upper sideband power + Lower sideband power

$P_u = P_c m^2/4 + P_c m^2/4)$

$= P_c m^2/2 = (Ac^2/2)\, P_m$ → P_m - power in the message signal m(t)

unmodulated -carrier power is $Pc=Ac^2/2R$

$P_c = A_c^2/2$(Carrier power, R=1)

Power savings

Power saving $= (P_{am} - P_{DSB\,SC}) / P_{am}$

$$\boxed{\text{Power savings} = \frac{P_t - P_t'}{P_t} =}$$

$$\frac{P_c(1+m^2/2) - P_c(m^2/2)}{P_c(1+m^2/2)}.$$

$$= \frac{1}{1+m^2/2}$$

For m=1 Power saving of 66.7%

Bandwidth $= (f_c + W) - (f_c - W) = 2W$ Hz (twice the highest modulating frequency)

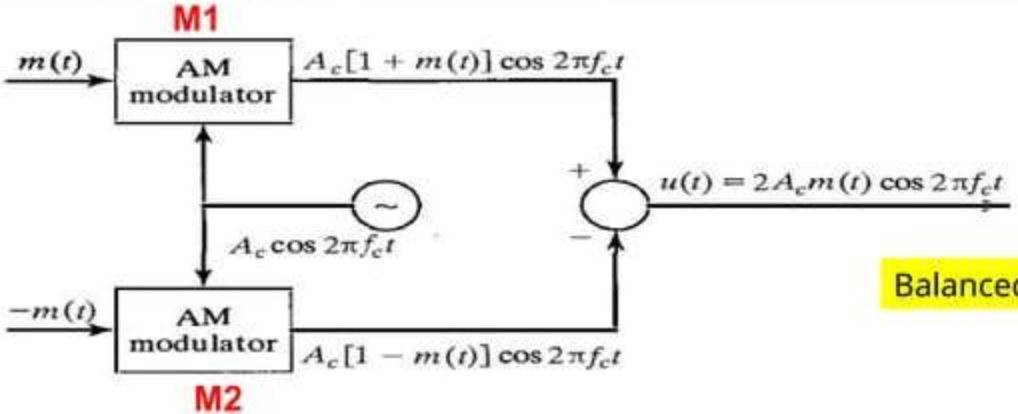
DSB-SC Modulators

1. Balanced Modulator
2. Ring Modulator

1. Balanced Modulator

- It is a simple method, consists of two identical AM
- two modulators are arranged in a balanced configuration in order to suppress the carrier signal.

- **Mixing operation** used to get **DSB-SC signal**
- **multiplication of message signal with carrier signal** is called a mixing operation
- mixer is basically a balanced modulator



Balanced modulator

It consists of two conventional AM modulators (square-law AM modulators)arranged in the balanced configuration so as to suppress the carrier

- modulators with approximately identical characteristics are selected so that the carrier component cancels out at the summing junction

The message signal m(t) is applied as it is to M1 and its inverted version -m(t) is applied to M2.
At the outputs of modulators M1 and M2 , we get standard AM signals s1(t) and s2(t) as under :

<u>Output of M1</u>

$s1(t) = A_c [1+m(t)] \cos(2\pi f_c t)$

Output of M2

$s2(t) = A_c[1 - m(t)]\cos(2\pi f_c t)$

Subtracrtor output

$$u(t) = s1(t) - s1(t)$$

$$= A_c[1+m(t)]\cos(2\pi f_c t)] - A_c[1 - m(t)]\cos(2\pi f_c t)$$

$$= A_c\cos(2\pi f_c t) + A_c m(t)\cos(2\pi f_c t) - [A_c\cos(2\pi f_c t) - A_c m(t)\cos(2\pi f_c t)\ m(t)]$$

$$= A_c\cos(2\pi f_c t) + A_c m(t)\cos(2\pi f_c t)$$

$$U(t) = 2 A_c m(t)\cos(2\pi f_c t)$$

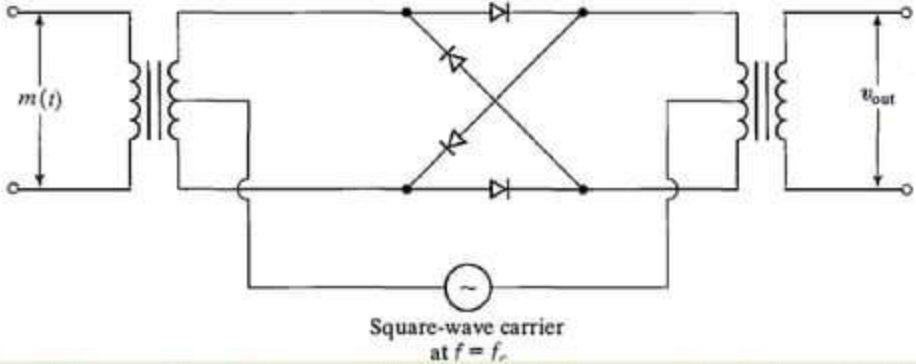
Hence, it represents a DSB-SC signal .

2. Ring Modulator(double-balanced mixer)

ring modulation is an implementation of frequency mixing, in which carrier signal is combined with message signal to yield a DSB-SC signal.

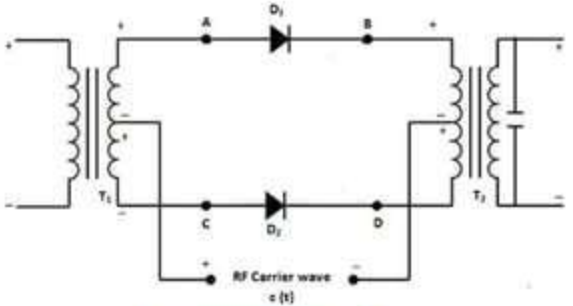
four diodes are connected in the ring structure.

switching of the diodes is controlled by c(t) (square wave of frequency f_c)
c(t) is applied to the center taps of the two transformers.

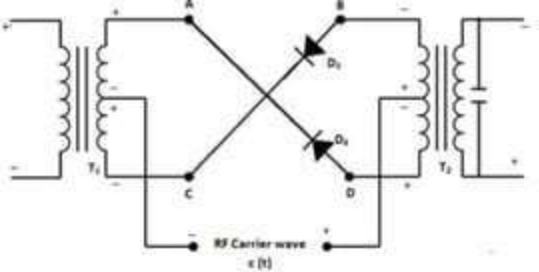


Ring Modulator

- **When c(t) >0**, top and bottom diodes conduct, two diodes in the cross arms are off.
  message signal m (t) is multiplied by + I .
- **When c(t) < 0**, the diodes in the cross arms of the ring conduct, the other two diodes are switched off



When c(t) >0



When c(t) < 0

operation of the ring modulator is represented by represented by the output voltage $v_0(t)$ $\boxed{v_0(t) = m(t)c(t)}$

c(t) is a periodic function, Fourier series of c(t) is

$$c(t) = \frac{4}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}\cos[2\pi f_c(2n-1)t]$$

ring modulator output is

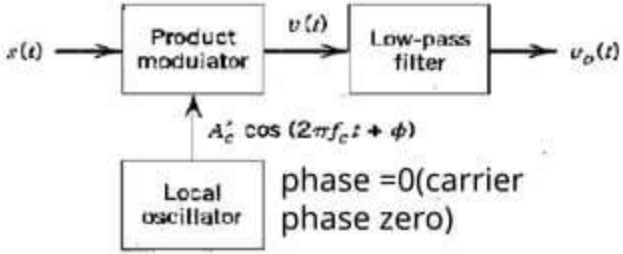
$$c(t)m(t) = \frac{4}{\pi} + \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{2n-1}\cos[2\pi f_c t(2n-1)]m(t)$$

$v_0(t)$ is passes through a bandpass filter with the center frequency f_c and the bandwidth 2W to get the desired DSB-SC AM signal

Demodulation of DSB SC

Coherent detection

(phase-coherent or synchronous demoulator)



phase =0(carrier phase zero)

During demodulation the local oscillator carrier is exactly coherent or synchronized, in both frequency and phase, with the carrier wave used to generate the modulated wave. This method of demodulation is known as **Coherent detection or synchronous demodulation**

Let local oscillator carrier = $A'_c \cos(2\pi f_c t + \phi)$

cosAcosB=1/2[cos(A+B)+cos(A-B)]

Output of product modulator is

$$
\begin{aligned}
v(t) &= A'_c \cos(2\pi f_c t + \phi) s(t) \\
&= A_c A'_c \cos(2\pi f_c t) \cos(2\pi f_c t + \phi) m(t) \\
&= \frac{1}{2} A_c A'_c \cos(4\pi f_c t + \phi) m(t) + \frac{1}{2} A_c A'_c \cos\phi \, m(t)
\end{aligned}
$$

DSB-SC modulated signal with carrier frequency $2f_c$

Proportional to baseband signal m(t)

Baseband signal m(t) lies in the interval $-W \le f_c \le W$

First term is removed by LPF with cutoff frequency f_c

$W < f_c < 2f_c - W$

Filter output is

$$v_o(t) = \frac{1}{2} A_c A_c' \cos\phi \, m(t)$$

Output $v_0(t)$ is propositional to m(t) when phase error ϕ is constant

When $\phi = 0$. amplitude of demodulated signal is maximum

When $\phi = \pm\pi/2$, amplitude of demodulated signal is minimum

When $\phi = 45°$, , amplitude of demodulated signal is reduced by $\sqrt{2}$

zero demodulated signal occurs when $\phi = \pm\pi/2$, represents the **quadrature null effect of the coherent detector**

Phase error ϕ causes the detector output attenuated by $\cos\phi$

As long as ϕ is constant, detector provide undistorted version of m(t)

Disadvantage of DSB-SC AM

channel bandwidth $B_c = 2$ W Hz for transmission

two sidebands are redundant

Single-Sideband AM

- Single-sideband suppressed carrier (SSBSC) or single sideband (SSB) is a form of amplitude modulation in which the carrier is fully suppressed and one of the sidebands (lower or upper) is also suppressed

  - Two sidebands of an AM signal are mirror images of each other,
  - Only one sideband is enough for transmission

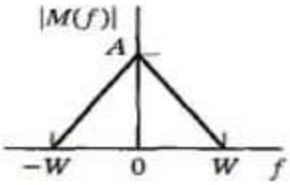
In SSB , Only the upper or lower sidebands are transmitted bandwidth efficient

SSB AM reduce the bandwidth of the transmitted signal to that of the baseband message signal m(t)

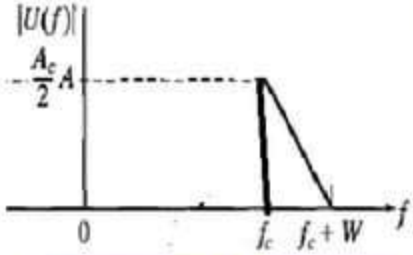
Spectrum of the SSB

$$U(f) = \frac{A_c}{2}[M(f \pm f_c)]$$

Spectrum consists of either upper or lower side band



spectra of m(t)



spectra of SSB signal

in U(f) at f = f_c, since the carrier is absent.

Power Content of SSB Signals

$$P_{SSB} = P_c\, m^2/4$$

$$P_c = A_c^2/2 \text{(Carrier power)}$$

Bandwidth = W

Power savings (compared to AM)

$$\text{Power saving} = (P_{am} - P_{SSB}) / P_{am}$$

$$= [P_c(1+m^2/2) - P_c\, m^2/4] / P_c(1+m^2/2)$$

$$= [(1+m^2/2) - m^2/4] / (1+m^2/2)$$

For m=1, $= [(1+1/2) - 1/4] / (1+1/2) = 5/6$

Power saving = 83.3 %

Power savings (compared to DSB)

$$\text{Power saving} = (P_{DSB} - P_{SSB}) / P_{DSB}$$

$$= [P_c\, m^2/2) - P_c\, m^2/4] / P_c\, m^2/2)$$

$$= [(m^2/2) - m^2/4] / (m^2/2)$$

For m=1, $= [1/2 - 1/4] / (1/2)$

Power saving = 50 %

Advantages of SSB

1. Bandwidth is half that of DSB-SC. Therefore twice the number of channels can be accommodated at a given frequency spectrum.
2. No carrier is transmitted, hence interference with other channel are avoided

Disadvantages of SSB

1. Transmission and reception of SSB is complex
2. Need high cost receiver

Applications of SSB

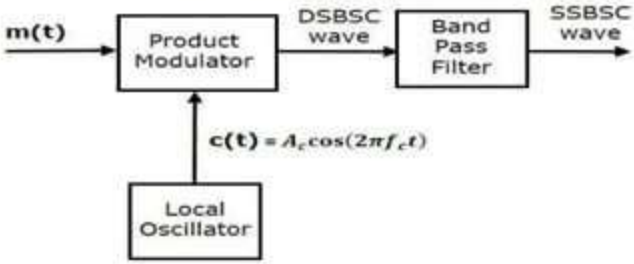
1. Wireless communication
2. Point to point communication
3. VHF and UHF communication systems

SSB generation

Methods

1. Frequency discrimination method(filter method or Balanced modulator)
2. Phase discrimination method
   a. phase shift method
   b. Weavers method(modified phase shift method)

1.Filter Method



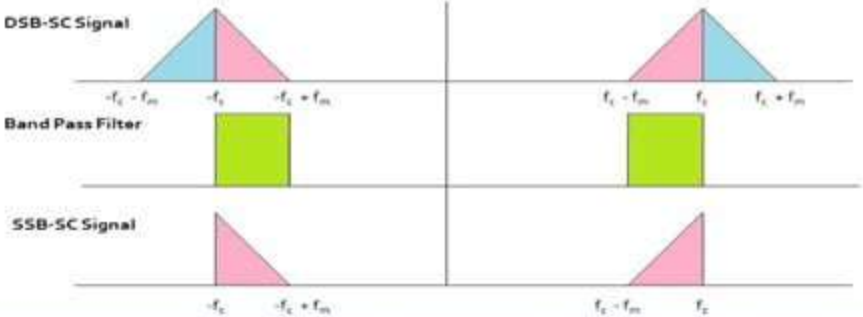
Filter Method

- Simple method
- The circuit consists of balanced modulator and the bandpass filter(sideband-suppression filter)
- The filter has flat bandpass and extremely high attenuation outside the bandpass.
- after the balanced modulator the unwanted sideband is removed by the bandpass filter

The filter method can be used for generating the SSB modulated wave if the message signal satisfies the following conditions :

- The message signal should not have any low frequency content . The audio signal posses this property
- The highest frequency in the spectrum of the message signal i.e. W Hz should be much smaller than carrier frequency fc .

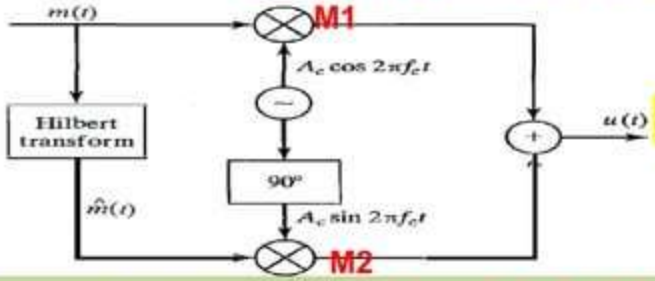
- The generation of a SSB signal is straightforward.
  - First, generate a double-sideband signal
  - Then apply an ideal pass-band filter to the result with cutoff frequencies of f_c and $f_c + W$ (or $f_c - W$) for the upper sideband (or lower sideband).
  - Practically, the approximate construction of an ideal filter is very difficult.



Disadvantage

1.frequency difference between the highest frequency in LSB and the lowest frequency in USB is too small . Therefore, design of the bandpass filter extremely difficult

2. Phase shift method



Generation of a SSB-AM

Phase shift method uses a **phase shift technique** that **causes one of the side bands to be canceled out**.

This method consists **of two balanced modulators** M_1 and M_2 and two 90° phase shifting network

message signal m(t) is applied to the product modulator M_1

And through Hilbert transform(provide 90° phase shift) m(t) is applied to the product modulator M_2.

$\hat{m}(t)$ is the Hilbert transform of m(t)

plus or minus sign determines the sideband
plus sign –lower sideband
minus sign upper sideband

Hilbert transform is viewed as a linear filter with impulse response h(t) = 1 /πt and frequency response

$$H(f) = \begin{cases} -j, & f > 0 \\ j, & f < 0 \\ 0, & f = 0 \end{cases}$$

Output of M1

$v1(t) = A_c m(t) \cos(2\pi f_c t) = A_c A_m \cos(2\pi f_m t) \cos(2\pi f_c t)$

$= (1/2) A_c A_m [\cos(2\pi f_m t + 2\pi f_c t) + \cos(2\pi f_m t - 2\pi f_c t)]$ - - - - -(1

Output of M2

$v2(t) = A_c m(t)\hat{} \sin(2\pi f_c t) = A_c A_m \sin(2\pi f_m t) \sin(2\pi f_c t)$

$= (1/2) A_c A_m [\cos(2\pi f_m t - 2\pi f_c t) - \cos(2\pi f_m t + 2\pi f_c t)]$ - - - - -(2

The outputs of M_1 and M_2 are applied to an adder .

Output of the adder is

$u(t) = v1(t) + v2(t)$

$= (1/2) A_c A_m [\cos(2\pi f_m t + 2\pi f_c t) + \cos(2\pi f_m t - 2\pi f_c t)] +$

$(1/2) A_c A_m [\cos(2\pi f_m t - 2\pi f_c t) - \cos(2\pi f_m t + 2\pi f_c t)]$

$= A_c A_m \cos(2\pi f_m t - 2\pi f_c t)$

Only one side band is present

single-sideband (SSB) AM signal is represented by

$$u(t) = A_c m(t) \cos 2\pi f_c t \mp A_c \hat{m}(t) \sin 2\pi f_c t$$

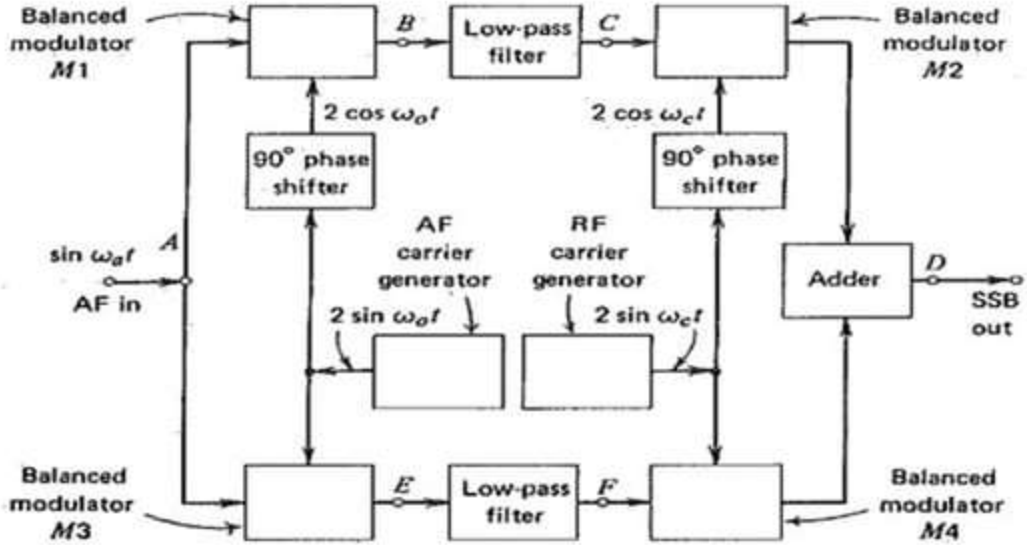
Advantages

- generate the SSB signal at any frequency, so the frequency up converter stage is not required .
- It can use the low audio frequencies as modulating signal
- It is easy to switch from one sideband to other .

Disadvantages

- design of the 90° phase shifting network for the modulating signal is extremely critical .
- This network has to provide a correct phase shift of 90° at all the modulating frequencies which is practically difficult to achieve .

3. Modified Phase shift method



- Third method of generating SSB was developed by Weaver.

- Design and implementation of accurate phase- shift networks was a difficult task.
- Weaver method eliminates the need for such networks, and replaces them with a pair of identical low-pass filters

- retaining the advantages of the phase-shift method, such as its ability to generate SSB at any frequency and use low audio frequencies
- use low audio frequencies without the need of AF phase-shift network to operate over a large range of audio frequencies.

there are a pair of parallel chains, each containing two oscillators, a low-pass filter, and two multipliers.

The first oscillator is low frequency oscillator set to the audio mid-point frequency f_o
The second is a high frequency (rf) oscillator at a frequency f_c. This is used for tuning the SSB output to the desired frequency f_c

- Instead of trying to phase-shift the whole range of **audio frequencies**, this method **uses an AF carrier f_o**, which is a fixed frequency in the middle of the audio band, 1650 Hz.
- A **phase shift is then applied to this frequency f_o only**, and the resulting voltages have been applied to the first pair of balanced modulators M1 and M3
- the **low-pass filters cut-off frequency is f_o** ,so that the input to the last pair of balanced modulators provide sideband suppression.

- **all lower sideband signals will be cancelled** regardless of whether audio frequencies are above or below f_o.
- If a **lower sideband signal is required**, the phase of the **carrier voltage applied to M1 may be changed by 180°**.

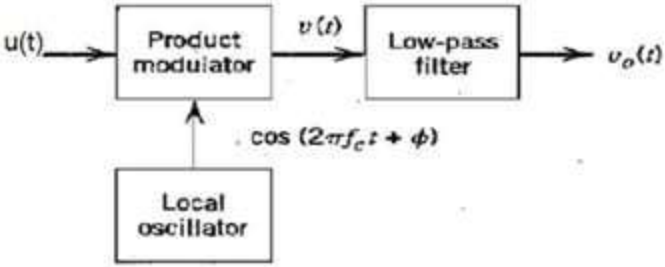
Advantages:

- generate SSB at any frequency.
- It uses low audio frequencies, without the need of AF phase-shift network

Disadvantages:

- complex.
- It is expensive and cannot be used commercially.

SSB demodulator



Coherent detection

During demodulation the local oscillator carrier is exactly coherent or synchronized, in both frequency and phase, with the carrier wave used to generate the modulated wave. This method of demodulation is known as **Coherent detection or synchronous demodulation**

Input of product modulator is

$$\boxed{u(t) = A_c m(t)\cos 2\pi f_c t \mp A_c \hat{m}(t)\sin 2\pi f_c t}$$

Output of product modulator is

$$\begin{aligned} v(t) &= u(t)\cos(2\pi f_c t + \phi) \\ &= \frac{1}{2}A_c m(t)\cos\phi + \frac{1}{2}A_c \hat{m}(t)\sin\phi \\ &\quad + \text{double-frequency terms.} \end{aligned}$$

Upper sideband

By passing the product signal ithrough an ideal lowpass filter, the double-frequency components are eliminated

$$V_o(t) = \frac{1}{2}A_c m(t)\cos\phi + \frac{1}{2}A_c \hat{m}(t)\sin\phi$$

phase offset(difference)

1.reduces the amplitude of the desired signal m(t) by cos ϕ,

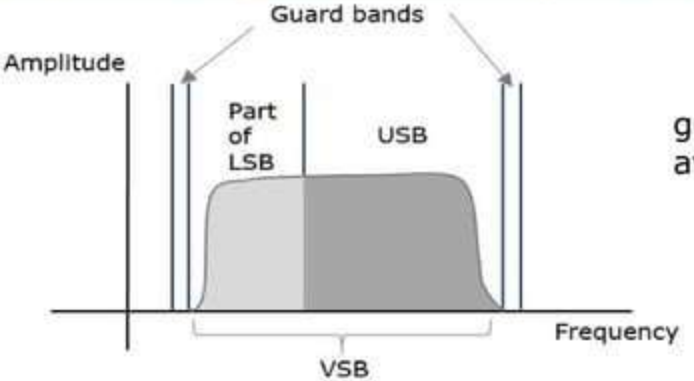
2. results in an undesirable sideband signal due to the presence of $\hat{m}(t)$

Vestigial-Sideband Modulation(VSB)

- Generation of SSB-SC is difficult due to the difficulty in isolating desired sideband.
- The required filter must have sharp cutoff characteristics when the baseband signal is of low frequency(Television and telegraph signals)
- As a result of which, some of the information may get lost.

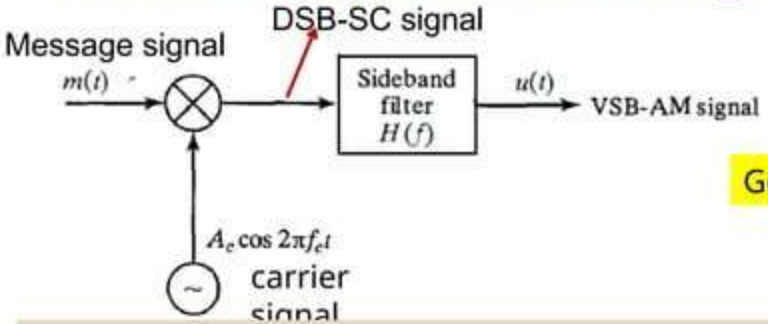
To avoid this loss, a technique is chosen, which is a compromise between **DSB-SC** and **SSB**, called as **Vestigial Sideband (VSB)** technique.

- In VSB, **desired sideband is allowed to pass completely and a small portion (trace or vestige) of the undesired sideband is allowed** , which compensate the loss of information
- A part of the signal called as vestige is modulated, along with one sideband.



guard band is very small to avoid the interferences.

Generation of VSB modulated signal –filter method



Generation of VSB-AM signal

generating a DSB-SC AM signal –Frequency discrimination method is used

a DSB-SC signal is generated and pass it through a sideband filter

Specially designed band pass filter with frequency response H(f)
H(f) is normalized so that $|H(f_c)| = 1/2$

In time domain, **VSB signal** is expressed as

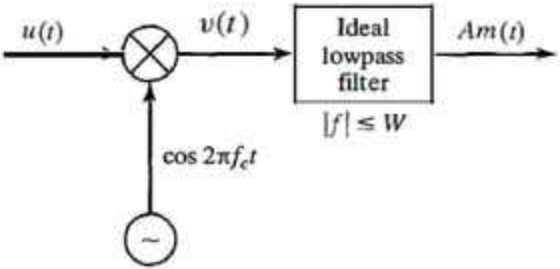
$$u(t) = [A_c m(t) \cos 2\pi f_c t] \star h(t)$$

h(t) is the impulse response of the VSB filter

In frequency domain

$$U(f) = \frac{A_c}{2}[M(f - f_c) + M(f + f_c)]H(f) \qquad \text{----(1)}$$

Demodulation of VSB signal



Demodulation of VSB signal

VSB signal u(t) is multiplied by the carrier component $\cos 2\pi f_c t$ and passes through an ideal lowpass filter

$$v(t) = u(t)\cos 2\pi f_c t$$

$$V(f) = \frac{1}{2}[U(f - f_c) + U(f + f_c)] \quad \text{----(2)}$$

Substituting (1) in (2)

$$V(f) = \frac{A_c}{4}[M(f - 2f_c) + M(f)]H(f - f_c)$$
$$+ \frac{A_c}{4}[M(f) + M(f + 2f_c)]H(f + f_c)$$

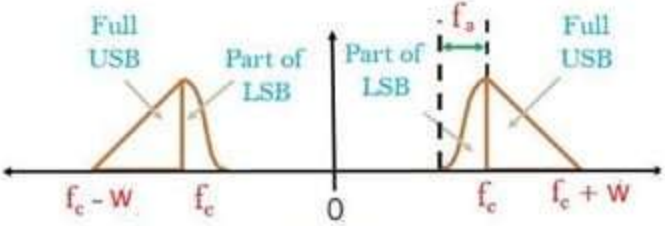
lowpass filter rejects the double-frequency terms and passes only the components in the frequency range $|f| \leq W$

signal spectrum at the output of the ideal lowpass filter is

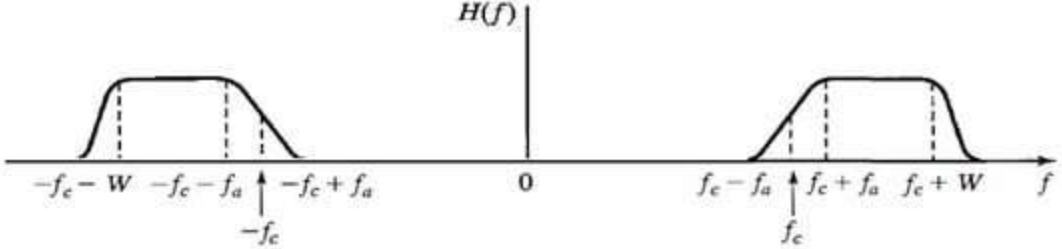
$$V_\ell(f) = \frac{A_c}{4} M(f)[H(f - f_c) + H(f + f_c)]$$

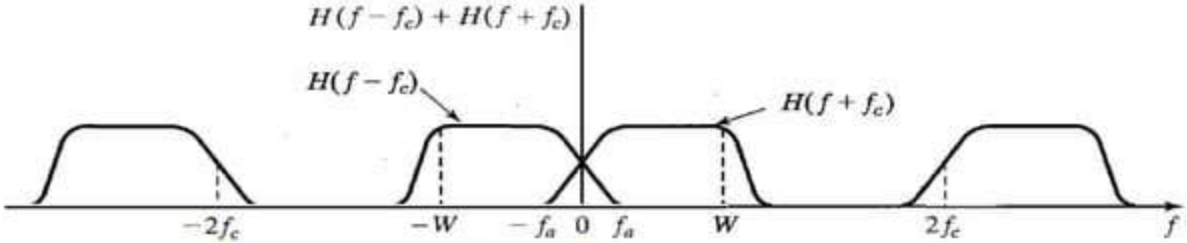
message signal at the output of the lowpass filter must be undistorted if .
VSB filter characteristic must satisfy the **condition**

$$\boxed{H(f - f_c) + H(f + f_c) = \text{constant} \quad |f| \leq W}$$

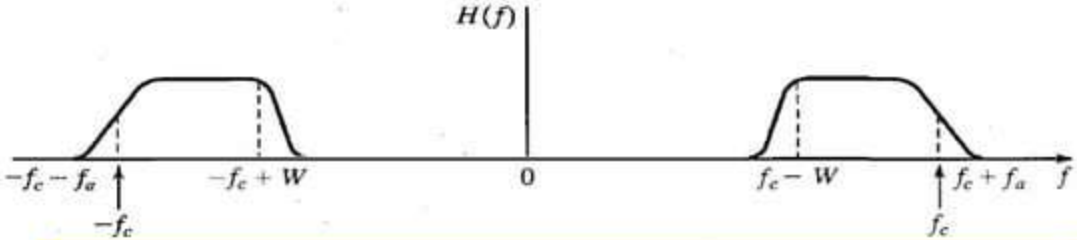


Spectrum of VSB signal





VSB filter characteristics



Frequency response of the VSB filter for selecting the lower sideband of the message signals

H(f) selects the upper sideband and a vestige of the lower sideband

Filter has odd symmetry about the carrier frequency fc in the frequency range $f_c - f_a \le |f| \le f_c + W$

f_a - conveniently selected frequency(small fraction of W)

Condition for reconstructing undistorted version of the transmitted signal

1. $f_a \ll W$

   f_a - conveniently selected frequency(small fraction of W)

2. VSB filter should have a linear phase over its pass band

   $(f_c - f_a \le |f| \le f_c + W)$

Bandwidth of VSB modulation

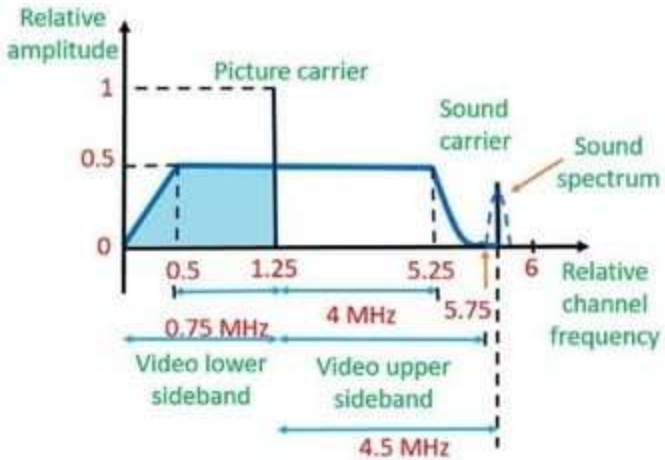
$$B_T = W + f_a$$

message bandwidth Width of the vestigial sideband

VSB modulation is used for TV transmission

- video signals required a bandwidth of at least 4 MHz
- For proper reception of television a minimum transmitted bandwidth of 9 MHz is required

-video signal has large bandwidth and low frequency
-need simple, inexpensive demodulation technique(envelope detector)



- VSB allows the transmission of upper sideband along with the vestige of lower sideband. Suppresses the remaining part.
- From 1.25 MHz of lower sideband band, 0.75 MHz vestige is transmitted and rest is suppressed.
- This basically simplifies the filtering requirements.

Hence, we can reduce the bandwidth requirement to 6 MHz from 9MHz.